Practice Test 3

Solutions and marking guide

20 main marks. Accept consistent transpose conventions and follow-through. Qualitative neural traces are not unique.

1 [3]

(a) $\hat{\mathbf{w}} = D\mathbf{a}_W$, with $D \in \mathbb{R}^{2\times N}$, readout in $\mathbb{R}^2$: **1**, half equation, half shapes.

(b) Encode each input vector into its population; decode and transform with $I_2, -3I_2, 2I_2, -2I_2$; sum the four contributions into W's encoding/current; decode W's activity to the output vector. **1** for encoding and transforms; **1** for combination, final population and readout. No weights derivation required.

2 [5]

(a) Thresholds R/2 and R; zero rate below each threshold; shared assigned endpoint rate r_{max} at R and 2R. **1** for these features; **1** for labelled axes and increasing nonlinear curves with a whole-curve horizontal stretch: $a_{2R}(x) = a_R(x/2)$. No exact curvature required.

(b) RMSE$(2R)$ = 2RMSE(R). Same normalized activities and regularization mean identity targets, decoders and residuals scale with R. **0.5** law, **0.5** mechanism. This concerns the full rescaled domain.

(c) **1**: represented-input intercept and preferred endpoint's assigned maximum rate remain fixed under both sweeps. **0.5**: recalibration alters interior LIF shapes, hence the activity basis/decoder fit and its error. **0.5**: fixed endpoints alone give no monotonic RMSE law. Fixed-current reasoning is a different experiment; feasibility is assumed. Gain and bias generally change. Do not require directional shape predictions.

3 [3]

1 input: 0→1→0 at 1/4 and 3/4 s. **1** A probe: approximately 0→1→0 with rounded transitions. **1** B probe: approximately 1→-1→1, opposite transition directions and additional qualitative smoothing/lag from its connection. Both probes can fluctuate slightly. B starts near 1 because baseline is settled. Do not demand exact settling by 1 s or a unique noisy trace.

4 [7]

(a) $A = 0_{d\times d}, B = I_d$; $A' = I_d, B' = \tau_r I_d = (0.05\,\mathrm{s})I_d$. **1** for A,B and **1** for A′,B′; reserve half of each mark for compatible d×d shapes and correct time scaling respectively. General recipe plus integrator values accepted. The 5 ms input filter does not set B′ here.

(b)

$$x(t) = \begin{cases} 0, & 0 \le t < 1/4, \\ t - 1/4, & 1/4 \le t < 3/4, \\ 1/2, & t \ge 3/4. \end{cases}$$

1.5, half per correct case including boundaries. Figure: **1** ideal continuous ramp from 0 to 1/2, then hold; **0.5** plausible neural approximation with small fluctuations and/or onset/offset transients. Align with Q3 input; no second input drawing earns or is needed for credit. Do not mandate a drift sign, exact error or saturation; mismatch need not produce a simple delay.

(c) **0.5** reduced spike-related fluctuations. **0.75** static decoding/representation error persists in rate mode, perturbs recurrent feedback and can cause drift/state-dependent hold error. **0.75** unequal input/recurrent filters change transients, violating the matched-filter construction. Probe filtering is an acceptable distinct residual source, but affects the displayed trace rather than creating persistent state drift. Two versions of one cause earn one cause's credit.

5 [2]

$F(x, c) = x + \tau cx$: **1.5**, split **0.75** identity feedback and **0.75** correctly scaled variable product. Input transform τ: **0.5**. Joint representation permits the nonlinear product. An input gate $\dot{x} = cu$ is a different system. No behavior comparison or trajectory required.